

Rifaximin in cirrhosis: Is its microbiological spotless record under threat?

Turner, A.M., Li, L., Monk, I.R. *et al.* Rifaximin prophylaxis causes resistance to the last-resort antibiotic daptomycin. *Nature* (2024).

Background and context

Patients with cirrhosis are particularly vulnerable to infections and other complications due to immune dysfunction and intestinal barrier impairment, leading to frequent antibiotic use and an associated rise in multidrug-resistant (MDR) organisms in this population.^{1–3} Among the antimicrobials frequently employed, rifaximin, a gut-specific antibiotic with minimal systemic absorption, has gained prominence for the prevention of recurrent hepatic encephalopathy (HE).^{4–6} Its approval in 2010 was based on compelling evidence of efficacy in reducing HE episodes, as demonstrated in the landmark randomized-controlled trial (RCT) by Bass *et al.*,⁷ which highlighted its ability to maintain HE remission in patients with cirrhosis when used alongside lactulose.

The risk of rifaximin inducing systemic antibiotic resistance is generally considered to be low because of its minimal absorption and targeted action on gut flora. However, its long-term use has raised concerns about the potential for selective pressure contributing to resistance in gut microbes. Some studies have shown that rifaximin may, to some extent, impact the abundance of antibiotic resistance genes in coagulase-negative staphylococci and *Clostridioides difficile*.^{8,9} The mechanisms by which the use of rifaximin might contribute to resistance are under investigation but clinical data confirming a direct causal link remain sparse.¹⁰

Enterococcus species, particularly vancomycin-resistant Enterococci (VRE), have emerged as significant pathogens in cirrhosis, associated with increased morbidity and mortality. Daptomycin-resistant VRE strains, although uncommon, represent a severe clinical challenge with limited therapeutic options.^{11,12} However, mechanisms inducing resistance to one of the last resort antibiotics are still not fully understood.

Objectives, methods, and findings

The recently published study in *Nature* by Turner *et al.*¹³ primarily aimed to uncover the mechanisms driving daptomycin resistance in vancomycin-resistant *Enterococcus faecium* (VREfm), focusing on unexplored genomic and phenotypic factors. Researchers analyzed a collection of 717 VREfm isolates from Australia, where they observed that 19.4% of the strains exhibited daptomycin resistance. To identify resistance

mechanisms, the team screened the genomes of DAP-resistant and susceptible strains. Known regulatory genes linked to daptomycin resistance showed no association, prompting an unbiased genome-wide association study across 998 isolates. Among 142 candidate mutations, the *rpoB* S491F mutation, located in the rifampicin-resistance-determining region of the RNA polymerase β subunit, which is a known locus for antibiotic resistance in enterococci, emerged as critical. Functional experiments confirmed that introducing the *rpoB* S491F mutation into daptomycin-susceptible strains induced resistance, while other mutations, such as T634 K, played compensatory roles by mitigating fitness deficits without directly driving resistance.¹³ The epidemiological significance of VREfm resistance to daptomycin was confirmed through a large, publicly available global dataset. Within this dataset, *rpoB* mutations were identified in approximately 15% of VREfm isolates associated with health-care environments. Notably, their emergence in 2006 and additional peak in 2010 coincided with the introduction of rifaximin to prevent HE, suggesting a potential link between rifaximin use and the development of *rpoB* mutations.¹³

Further investigation revealed that daptomycin resistance was strongly associated with rifaximin exposure. One cohort colonized with VREfm consisted of individuals who had been treated with rifaximin ($n = 96$), while the comparison cohort also colonized with VREfm included those who had not been exposed to rifaximin ($n = 116$). The majority of rifaximin-treated individuals had chronic liver disease with HE. Among this group, a significantly higher prevalence of *rpoB*-mediated daptomycin resistance in VREfm was observed compared to the non-rifaximin-treated cohort (odds ratio 8.69). The same observation was made in a small group of patients who had received rifaximin during hematopoietic stem-cell transplantation. This analysis indicated a clear association between rifaximin use and the presence of *rpoB* mutations in the context of VREfm colonization.¹³ Thereafter, findings were corroborated in a mouse colonization model, where rifaximin and rifampicin, but not daptomycin, consistently induced *rpoB*-mediated daptomycin resistance.¹³

Mechanistic studies highlighted how rifaximin-induced *rpoB* mutations alter bacterial cell membrane properties. Lipidomic analysis revealed reduced levels of anionic cardiolipins and phosphatidylglycerol, alongside increased levels of cationic lysyl-phosphatidylglycerol and digalactosyl-diacylglycerol, which collectively impair daptomycin membrane binding. Overexpression of the *prdRAB* operon in *rpoB*-mutant strains was identified as a key driver of these lipid changes. Disrupting *prdR* or its associated proteins restored daptomycin susceptibility, demonstrating its central role in resistance development.¹³



Significance of findings

This study highlights an emerging issue at the intersection of rifaximin use and the development of cross-resistance to a last resort antibiotic in VREfm via *rpoB* mutations. The findings present compelling experimental evidence about the potential mechanism but also raise significant unanswered questions, emphasizing the need for a nuanced evaluation of clinical implications and future directions in research.

The association between rifaximin exposure and *rpoB*-mediated daptomycin resistance is particularly concerning for patients with cirrhosis, a population at risk of being colonized with VREfm and commonly treated with rifaximin for HE.^{3,14} However, the relationship between rifaximin exposure duration and the risk of developing resistance, as well as the influence of additional factors such as concurrent bacterial infections requiring antibiotic therapy, remains uncertain. Understanding how these variables interact is crucial for refining treatment strategies and mitigating the risk of resistance. To better inform clinical decision-making and tailor treatment regimens, prospective studies and a re-evaluation of already performed RCTs are needed to more clearly define this relationship and identify critical factors that may predispose patients to developing daptomycin resistance. The retrospective design of this study introduces inherent limitations, such as potential confounding factors and biases. If we accept as proven that rifaximin, in this clinical context and with specific treatment modalities, serves as a potent inducer of resistance, the lack of reports from existing RCTs evaluating rifaximin for HE on the emergence of daptomycin resistance suggests that this phenomenon may have been overlooked or underreported.^{5,7,15}

It remains puzzling why, in this study, rifaximin-induced resistance appears specifically focused on VREfm. From the mechanistic point of view, rifaximin has the potential to induce resistance in vancomycin-susceptible strains through *rpoB* mutations. The study's focus on VREfm is likely due to the significant clinical impact of dual resistance to both vancomycin and daptomycin, which can greatly complicate treatment options and worsen patient outcomes.¹⁶ This specificity highlights the need to further investigate whether resistance mechanisms induced by rifaximin are universally applicable or if they are dependent on the microbiological context, such as pre-existing vancomycin resistance, and whether these mechanisms also apply to other enterococcal species or different bacterial pathogens.

Moreover, the apparent discrepancy between the widespread use of rifampicin, a rifamycin derivative like rifaximin, and the rarity of reported daptomycin resistance is intriguing, especially since *in vitro* studies showed that rifampicin has the same potential as rifaximin to induce daptomycin cross-resistance.¹³ Rifampicin, commonly used in tuberculosis and endocarditis treatment, is often part of combination regimens. Experimental studies indicated that such combinations, including rifampicin with daptomycin, can enhance susceptibility even in resistant strains.¹⁷ This may partially explain why rifaximin, typically used as monotherapy in cirrhosis, poses a unique risk of resistance. Geographic variations in VRE treatment standards can additionally affect the clinical relevance of daptomycin resistance. Regions prioritizing linezolid for VRE may see less impact from daptomycin resistance. However, the risk of dual resistance to linezolid and daptomycin in these settings remains a

concerning possibility, highlighting the importance of balancing antibiotic stewardship with effective treatment strategies. Despite these concerns, rifaximin's proven efficacy in preventing recurrent HE must be weighed carefully against the risks. While current evidence highlights a distinct and significant resistance mechanism, the clinical relevance of this finding is not yet compelling enough to justify immediate and fundamental changes in clinical practice. Ongoing surveillance and collection of additional high-quality data is critical to addressing these gaps and ensuring rifaximin's continued safety and efficacy in cirrhosis management.

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Conflict of interest

The author of this study declares that they do not have any conflict of interest. Please refer to the accompanying ICMJE disclosure form for further details.

Supplementary data

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